Assignment - 10

Q.2.

$I_{max} = 0.6 \times D$

$\Rightarrow \boxed{D = \frac{36}{0.6} = 60\%}$

Q.3.

$I = 0.6 \times D$

$\Rightarrow 48 = 0.6 \times D$

$D = 80\%$

$\Rightarrow T_s = \frac{1}{1000} = 10^{-3}$

$\Rightarrow$ Positive level time $= DT_s = 0.8 \times 10^{-3}$

$= \boxed{800\ \mu s}$

Q.4.

$$V_{in} = \frac{5 \times R_{eq}}{R_{eq} + 330}$$

$$R_{eq} = \frac{4.7k \times 300}{4.7k + 300} = 282$$

$\Rightarrow V_{in} = \frac{5 \times 282}{282 + 330} = 2.303$

$\boxed{V_{in} = 2.30\ V}$

Q.5.

$$V_{CP-PE} = \frac{12 \times R_{eq}}{R_{eq} + 1000}$$

$$R_{eq} = \frac{2740 \times 1300}{2740 + 1300} = 881.68$$

$$\Rightarrow V_{CP-PE} = \frac{12 \times 881.68}{881.68 + 1000}$$

$$\boxed{V_{CP-PE} = 5.622\ V}$$